

Package ‘RERconverge’

April 13, 2026

Type Package

Title Analysis of Convergence Between Organismal Traits and DNA/Protein Sequences

Version 0.3.0

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Description RERConverge is a set of software written in R that estimates the correlation between relative evolutionary rates of genes and the evolution of a convergent binary or continuous trait across a phylogeny.

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Encoding UTF-8

LazyData true

RoxygenNote 7.3.2

Depends RColorBrewer, gplots, phytools, geiger, knitr, Rcpp, RcppArmadillo, weights, phangorn, ggplot2, dplyr, FSA, castor, Matrix, TreeTools, progress, impute, ggtree

Imports Rcpp (>= 0.11.0), ape

LinkingTo Rcpp, RcppArmadillo, ape

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adaptivePermutation	<i>Adaptively calculates permutation p-value for a single element</i>
---------------------	---

Description

Adaptively calculates permutation p-value for a single element

Usage

```
adaptivePermutation(
  rer,
  permulated_trees,
  observed_stats,
  alpha = 0.05,
  mode = "binary"
)
```

Arguments

rer	a row matrix for the given element (e.g., a row from the matrix output from getAllResiduals)
permulated_trees	a list containing n sets of permulated trees (e.g., output from generatePermulatedBinPhen)
observed_stats	computed statistics for the observed trait (e.g., output from getAllCor , correlateWithBinaryPhenotype or correlateWithContinuousPhenotype)

alpha	the significance level to control (default = 0.05)
mode	"binary" for binary permutations, "continuous" for continuous permutations (default = "binary")

Value

out data frame containing permPval (permutation p-value) and the corrected score (negative signifies deceleration, positive signifies acceleration)

asymmRerootingMethod	<i>Calculates ancestral likelihoods of ancestral states for discrete characters</i>
----------------------	---

Description

Calculates ancestral likelihoods of ancestral states for discrete characters

Usage

```
asymmRerootingMethod(tree, tips, Q)
```

Arguments

tree	An object of class "phylo"
tips	The phenotype data for the species at the tips of the tree. Must be in the same order as tree\$tip.label
Q	A transition rate matrix fit on the tree and tips

Value

A matrix of ancestral likelihoods with rows in the same order as the internal nodes in the tree

boxPlotTest	<i>Compares prediction accuracy under different rate models and represents the results as a series of boxplots</i>
-------------	--

Description

Compares prediction accuracy under different rate models and represents the results as a series of boxplots

Usage

```
boxPlotTest(
  treesObj,
  phenvals,
  rate_models,
  nsims = 100,
  confidence_threshold = NULL,
  ...
)
```

Arguments

treesObj	the trees object returned by readTrees
phenvals	the named phenotype vector with names matching the tip labels in the trees object
rate_models	a list of rate models to compare
nsims	the number of simulations to perform for each model of evolution (each rate model)
confidence_threshold	if not NULL, only compares prediction accuracy for nodes whose maximum likelihood is greater than the confidence threshold. Must be between 0 and 1.

Value

a list of boxplots for each rate model

calculateCorPermuted	<i>Calculate permutation correlation statistics</i>
----------------------	---

Description

Calculate permutation correlation statistics

Usage

```
calculateCorPermuted(
  permulated.paths,
  RERmat,
  min.sp = 10,
  min.pos = 2,
  method = "k"
)
```

Arguments

permulated.paths	A nested list of permulated paths (e.g., output of calculatePermulatedPaths_apply)
RERmat	An RER matrix calculated using getAllResiduals .
min.sp	Minimum number of species that must be present for a gene
min.pos	Minimum number of species that must be present in the foreground (non-zero phenotype values)
method	Method used to compute correlations. Accepts the same arguments as cor . Set to "auto" to select automatically based on the number of unique values in charP. This will also auto set the winsorization for Pearson correlation. Set winsorize-trait=some number and winsorizeRER=some number to override

Value

A nested list containing the correlation statistics for the permutations

calculatePermulatedPaths_apply

Calculates the paths for all permulated trees for a list of genes

Description

Calculates the paths for all permulated trees for a list of genes

Usage

```
calculatePermulatedPaths_apply(permulated.trees.list, map.list, treesObj)
```

Arguments

permulated.trees.list	A nested list of permulated phenotype trees for all the genes. The top layer lists the genes, and the nested layer lists the permulated binary trees (phylo objects) for each gene
map.list	A list of maps corresponding to the genes listed in permulated.trees.list, the output of matchAllNodesClades
treesObj	treesObj from readTrees

Value

permulated.paths.list A nested list of permulated paths corresponding to the nested list of permulated phenotype trees

`categoricalPermutations`*Runs permutations for categorical data*

Description

Runs permutations for categorical data

Usage

```
categoricalPermutations(  
  treesObj,  
  phenvals,  
  rm,  
  rp = "auto",  
  ntrees,  
  percent_relax = 0  
)
```

Arguments

<code>treesObj</code>	trees object returned by <code>readTrees</code>
<code>phenvals</code>	the named phenotype vector
<code>rm</code>	the rate model, it should be the same as the one used to reconstruct the ancestral history of the trait
<code>rp</code>	root prior, the default is <code>auto</code>
<code>ntrees</code>	the number of null trees to generate
<code>root_prob</code>	The probabilities of the different states at the root for the simulations. Can be "flat", "stationary", or a numeric vector of length <code>Nstates</code> . See the <code>root_probabilities</code> parameter under <code>simulate_mk_model</code> in the <code>castor</code> package for more information.

Value

a set of permulated phenotype trees

char2Paths	<i>turns a named vector of characters into a paths vector to be used with getAllCor</i>
------------	---

Description

turns a named vector of characters into a paths vector to be used with [getAllCor](#)

Usage

```
char2Paths(  
  tip.vals,  
  treesObj,  
  altMasterTree = NULL,  
  metric = "diff",  
  se.filter = -1,  
  ...  
)
```

Arguments

tip.vals	the trait/phenotype/character value at the tip, names(tip.vals) should match some of the mastertree\$tip.label, though a perfect match is not required
treesObj	A treesObj created by readTrees
metric	The metric used to translate node values into branch values. "Diff" takes the difference and makes the result phylogenetically independent. Other possible values are "mean" (the mean of the two values) and "last" the value of the most recent species on the branch. The last two options are not phylogenetically independent and downstream computations for those are not yet implemented
se.filter	Will remove branch values that are not at least se.filter*edge.se away from 0 (where edge.se is the standard error in the estimate for the edge value). Only implemented for metric="diff". Set se.filter to a positive value to filter. By default no filtering is done.

Value

A vector of length equal to the number of paths in treesObj

`char2PathsCategorical` turns a named vector of characters into a paths vector to be used with [getAllCor](#) for categorical traits

Description

turns a named vector of characters into a paths vector to be used with [getAllCor](#) for categorical traits

Usage

```
char2PathsCategorical(  
  tipvals,  
  treesObj,  
  useSpecies = NULL,  
  model = "ER",  
  plot = FALSE,  
  anctrail = NULL,  
  root_prior = "auto"  
)
```

Arguments

<code>tipvals</code>	the trait/phenotype/character value at the tip, <code>names(tip.vals)</code> should match some of the <code>masterTree\$tip.label</code> , though a perfect match is not required
<code>treesObj</code>	A <code>treesObj</code> created by readTrees
<code>useSpecies</code>	Give only a subset of the species to use for ancestral state reconstruction (e.g., only those species for which the trait can be reliably determined).
<code>model</code>	Specifies what rate model to use
<code>plot</code>	Plots a phenotype tree
<code>anctrail</code>	The trait to use for all ancestral species instead of inferring ancestral states if not NULL. The default is NULL.
<code>root_prior</code>	The prior probabilities of each trait at the root used to fit the transition matrix. Can be a vector of length equal to the number of states or one of the following: "flat", "empirical", "stationary", "likelihoods", "max_likelihood".

Value

A vector of length equal to the number of paths in `treesObj`

char2TreeCategorical *Creates a categorical trait tree from a set of tip species.*

Description

Creates a categorical trait tree from a set of tip species.

Usage

```
char2TreeCategorical(
  tipvals,
  treesObj,
  useSpecies = NULL,
  model = "ER",
  root_prior = "auto",
  plot = FALSE,
  anctrail = NULL
)
```

Arguments

tipvals	the trait/phenotype/character value at the tip, names(tip.vals) should match some of the mastertree\$tip.label, though a perfect match is not required
treesObj	A treesObj created by readTrees
useSpecies	Give only a subset of the species to use for ancestral state reconstruction (e.g., only those species for which the trait can be reliably determined).
model	Specifies what rate model to use
root_prior	The prior probabilities of each trait at the root used to fit the transition matrix. Can be a vector of length equal to the number of states or one of the following: "flat", "empirical", "stationary", "likelihoods", "max_likelihood".
plot	Plots a phenotype tree
anctrail	The trait to use for all ancestral species instead of inferring ancestral states if not NULL. The default is NULL.

Value

A tree with edge.length representing phenotype states

click_select_foreground_branches

Interactive click-based function to select foreground branches showing convergent binary trait

Description

Interactive click-based function to select foreground branches showing convergent binary trait

Usage

```
click_select_foreground_branches(mastertre)
```

Arguments

mastertre. A phylo tree object defining the topology of all species

Value

A binary trait tree with branch lengths of 1 for selected foreground species and lengths of 0 for the rest

combinePermData

Combines batches of permutations

Description

Combines batches of permutations

Usage

```
combinePermData(permdat1, permdat2, enrich = T)
```

Arguments

permdat1 Batch of permutations output from [getPermsContinuous](#)
 permdat2 Batch of permutations output from [getPermsContinuous](#)
 enrich Default T. Specifies if both 'permdat1' and 'permdat2' contain enrichment permutations

Value

Combined permutations

compareRateModels	<i>Performs pairwise likelihood ratio comparisons on a list of rate models</i>
-------------------	--

Description

Performs pairwise likelihood ratio comparisons on a list of rate models

Usage

```
compareRateModels(
  rate_models,
  treesObj,
  phenvals,
  nsims = 100,
  nested_only = FALSE,
  return_type = "pvals",
  ...
)
```

Arguments

rate_models	a named list of rate models in matrix form
treesObj	the trees object returned by readTrees
phenvals	the named phenotype vector with names matching the tip labels on the tree
nsims	the number of simulations to use in getLRnullDistributions when the rate models are not nested
nested_only	a boolean indicating whether comparisons should only be made between nested models. If TRUE, this decreases runtime because simulations to get the null distribution are not performed.
return_type	can be "pvals" or "ratios" indicating whether to return p values obtained from the likelihood ratios or the likelihood ratios themselves
...	additional parameters for the castor function, fit_mk used to fit the transition matrix

Value

a square matrix of p-values or likelihood ratios for pairwise comparisons between rate models

coreGetResiduals	<i>Get All Residuals core function</i>
------------------	--

Usage

```
coreGetResiduals(
  treesObj,
  nvMod = NULL,
  n.pcs = 0,
  cutoff = NULL,
  useSpecies = NULL,
  min.sp = 10,
  min.valid = 20,
  doOnly = NULL,
  maxT = NULL,
  block.do = F,
  weights = NULL,
  use.weights = T,
  interaction = F
)
```

Arguments

treesObj	A trees object from readTrees or readTrees2
nvMod	A normalization model with nrow(treesObj\$paths) rows and any number of columns (optional).
n.pcs	Number of principal components to normalize by (default: 0, mean normalization).
cutoff	A cutoff for branches to be ignored (optional, lowest 5 \itemuseSpeciesSpecies subset to use (optional). \itemmin.spMinimum species in a tree (default: 10). \itemmin.validMinimum number of non NA values (after filtering) that must be present for regression to be computed (default: 20). \itemdoOnlyOnly do specific trees (optional). \itemmaxTMaximum number of trees to do (optional, mostly useful for debugging). \itemblock.doProcess rows in blocks (default: FALSE). \itemweightsWeights for calculations (optional, if treesObj has weights those are used). \itemuse.weightsCan be used to turn off weighted regression. \iteminteractionUse PC interactions when building the normalization model (default: FALSE).

A list containing calculated residuals

Calculate residuals for a given trees object using a set of normalization covariates

correlateWithBinaryPhenotype

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype paths vector for a binary phenotype made with [tree2Paths](#)

Description

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype paths vector for a binary phenotype made with [tree2Paths](#)

Usage

```
correlateWithBinaryPhenotype(
  RERmat,
  charP,
  min.sp = 10,
  min.pos = 2,
  weighted = "auto",
  winsorizeRER = NULL,
  winsorizetrait = NULL,
  bootstrap = F,
  bootn = 1000,
  sort = F
)
```

Arguments

RERmat	RER matrix returned by getAllResiduals
charP	Phenotype vector returned by tree2Paths or char2Paths
min.sp	Minimum number of species that must be present for a gene
min.pos	Minimum number of species that must be present in the foreground (non-zero phenotype values)
weighted	Perform weighted correlation. This option turns on weighted correlation that uses the weights computed by foreground2Tree (wholeClade=T). This setting will treat each clade a single observation for the purpose of p-value estimation. The function will guess automatically if the charP vector is of "weighted" type and there should be not need to set this parameter.
winsorizeRER	Winsorize RER values before computing Pearson correlation. winsorizeRER=3 will set the 3 most extreme RER values at each end of each row to the 4th most extreme value.
winsorizetrait	Winsorize phenotype vector values before computing Pearson correlation. winsorizeRER=3 will set the 3 most extreme RER values at each end of each row to the 4th most extreme value.
bootstrap	Toggle bootstrapping (for weighted pearson correlation)

bootn	Number of runs to use when bootstrapping. Will be ignored if bootstrap is false.
sort	If set to TRUE, sorts rows by $\log_{10}(\text{P-value}) * -1 \setminus \text{sign}(\text{Rho})$

correlateWithCategoricalPhenotype

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype paths vector for a categorical phenotype made with [char2PathsCategorical](#)

Description

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype paths vector for a categorical phenotype made with [char2PathsCategorical](#)

Usage

```
correlateWithCategoricalPhenotype(
  RERmat,
  charP,
  min.sp = 10,
  min.pos = 2,
  method = "kw",
  winsorizeRER = NULL,
  winsorizetrait = NULL,
  sort = F
)
```

Arguments

RERmat	RER matrix returned by getAllResiduals
charP	Phenotype vector returned by tree2Paths or char2Paths
min.sp	Minimum number of species that must be present for a gene
min.pos	Minimum number of species that must be present in a category
method	Method used to compute correlations. Use "kw" to use Kruskal Wallis. Use "aov" to use ANOVA. It must be one of the strings "kw" or "aov".
winsorizeRER	Winsorize RER values before computing Pearson correlation. winsorizeRER=3 will set the 3 most extreme values at each end of each RER row to the value closest to 0.
winsorizetrait	Winsorize trait values before computing Pearson correlation. winsorizetrait=3 will set the 3 most extreme values of the trait values to the value closest to 0.
sort	If set to TRUE, sorts rows by $\log_{10}(\text{P-value}) * -1 \setminus \text{sign}(\text{Rho})$

Value

A list containing a list object with correlation values, p-values, and the number of data points used for each tree and a list of list objects for each pairwise test with correlation values and p-values.

correlateWithContinuousPhenotype

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype paths vector for a continuous phenotype made with [char2Paths](#)

Description

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype paths vector for a continuous phenotype made with [char2Paths](#)

Usage

```
correlateWithContinuousPhenotype(
  RERmat,
  charP,
  min.sp = 10,
  winsorizeRER = 3,
  winsorizetrait = 3,
  sort = F
)
```

Arguments

RERmat	RER matrix returned by getAllResiduals
charP	phenotype vector returned by tree2Paths or char2Paths
min.sp	Minimum number of species that must be present for a gene
winsorizeRER	Winsorize RER values before computing Pearson correlation. winsorizeRER=3 will set the 3 most extreme values at each end of each RER row to the the value closest to 0.
winsorizetrait	Winsorize trait values before computing Pearson correlation. winsorizetrait=3 will set the 3 most extreme values of the trait values to the value closest to 0.
sort	If set to TRUE, sorts rows by $\log_{10}(\text{P-value}) * -1 \setminus * \text{sign}(\text{Rho})$
method	Method used to compute correlations. Accepts the same arguments as cor . Set to "auto" to select automatically based on the number of unique values in charP. This will also auto set the winsorization for Pearson correlation. Set winsorize-trait=some number and winsorizeRER=some number to override

edgeVars	<i>maps a vector of traits onto a reference tree</i>
----------	--

Description

maps a vector of traits onto a reference tree

Usage

```
edgeVars(mastertree, tip.vals, metric = "diff", se.filter = -1, return.var = F)
```

Arguments

mastertree	the tree species the topology of the output tree and the branch lengths are used to infer ancestral states
tip.vals	the trait/phenotype/character value at the tip, names(tip.vals) should match some of the mastertree\$tip.label, though a perfect match is not required
metric	The metric used to translate node values into branch values. "Diff" takes the difference and makes the result phylogenetically independent. Other possible values are "mean" (the mean of the two values) and "last" the value of the most recent species on the branch. The last two options are not phylogenetically independent and downstream computations for those are not yet implemented
se.filter	Will remove branch values that are not at least se.filter*edge.se away from 0 (where edge.se is the standard error in the estimate for the edge value). Only implemented for metric="diff". Set se.filter to a positive value to filter. By default no filtering is done.
return.var	Returns the variance instead of the mean. Useful for seeing which estimates have high confidence.

Value

A phylo tree with branch values computed from the input tip.values

estimatePhangornTree	<i>Estimate a ML tree from a given topology Uses 'pml' and 'optim.pml' from the 'phangorn' package to estimate the tree.</i>
----------------------	--

Description

Estimate a ML tree from a given topology Uses 'pml' and 'optim.pml' from the 'phangorn' package to estimate the tree.

Usage

```
estimatePhangornTree(
  alnfile,
  treefile,
  submodel = "LG",
  type = "AA",
  format = "fasta",
  k = 4,
  ...
)
```

Arguments

alnfile	The path to the sequence alignment file
treefile	The path to the master tree file (whose topology will be used to generate the tree)
submodel	Substitution model to be used to estimate the tree
type	"AA" for amino acid or "DNA" for DNA
format	Format of the alignment file (commonly used formats include fasta and phylip)
k	Number of intervals in the discrete gamma distribution for 'pml'
...	Further arguments passed to 'pml' or 'optim.pml'

Value

A list: 'tree.opt' is the tree from the optimized output of 'optim.pml'; 'results.opt' is the optimized output of 'optim.pml'; 'results.init' is the initial results estimated by 'pml'

See Also

[phyDat](#) for alignment formats, [pml](#) and [optim.pml](#) for tree estimation

```
estimatePhangornTreeAll
```

*Estimates a ML trees from a given topology for a list of alignments.
Writes a text file compatible with readTrees(). Uses 'pml' and 'optim.pml' from the 'phangorn' package to estimate the tree.*

Description

Estimates a ML trees from a given topology for a list of alignments. Writes a text file compatible with readTrees(). Uses 'pml' and 'optim.pml' from the 'phangorn' package to estimate the tree.

Usage

```
estimatePhangornTreeAll(
  alnfiles = NULL,
  alndir = NULL,
  pattern = NULL,
  treefile,
  output.file = NULL,
  submodel = "LG",
  type = "AA",
  format = "fasta",
  k = 4,
  ...
)
```

Arguments

<code>alnfiles</code>	A character vector of paths to alignment files. Such as one produced by list.files . Either <code>alnfiles</code> or <code>alndir</code> must be supplied.
<code>alndir</code>	The path to directories with alignment files. Trees will be named with the alignment file name after stripping the extension. Either <code>alnfiles</code> or <code>alndir</code> must be supplied.
<code>pattern</code>	An optional regular expression for files in the <code>alndir</code> director. As in <code>".*fasta"</code> .
<code>treefile</code>	The path to the master tree file (whose topology will be used to generate the tree)
<code>output.file</code>	The file where the output trees will be written. This file can be read with <code>readTrees()</code> .
<code>submodel</code>	Substitution model to be used to estimate the tree
<code>type</code>	"AA" for amino acid or "DNA" for DNA
<code>format</code>	Format of the alignment file (commonly used formats include fasta and phylip)
<code>k</code>	Number of intervals in the discrete gamma distribution for 'pml'
<code>...</code>	Further arguments passed to 'pml' or 'optim.pml'

Value

A list: 'tree.opt' is the tree from the optimized output of 'optim.pml'; 'results.opt' is the optimized output of 'optim.pml'; 'results.init' is the initial results estimated by 'pml'

See Also

[phyDat](#) for alignment formats, [pml](#) and [optim.pml](#) for tree estimation

fastwilcoxGMTall	<i>Performs Wilcoxon Rank-Sum pathway enrichment on pathway gene-sets</i>
------------------	---

Description

Performs Wilcoxon Rank-Sum pathway enrichment on pathway genesets

Usage

```
fastwilcoxGMTall(vals, annotList, alternative = "two.sided", ...)
```

Arguments

vals	The output from getStat run on RERconverge correlation functions (correlateWithContinuousPhenotype, correlateWithBinaryPhenotype, getAllCor)
annotList	Pathway annotations formatted as a list of lists of annotation information

Value

A list of pathways with enrichment statistics

filterByCriteria	<i>Filters a list of rate models based on a boolean expression expressing filtering criteria</i>
------------------	--

Description

Filters a list of rate models based on a boolean expression expressing filtering criteria

Usage

```
filterByCriteria(models, criteria)
```

Arguments

models	a list of rate models
criteria	a function that is passed to supply specifying the desired criteria to filter by. It should take one argument, a rate model, and should be a boolean expression expressing criteria for the rate model.

Value

returns a subset of the list of rate models that match the criteria specified in the criteria function

fixPseudoroot	<i>pseudoroot trait tree to match the psuedoroot of trees from readTrees if the trees are reconcilable</i>
---------------	--

Description

pseudoroot trait tree to match the psuedoroot of trees from [readTrees](#) if the trees are reconcilable

Usage

```
fixPseudoroot(tree, treesObj)
```

Arguments

tree	A trait tree with branch lengths representing trait values
treesObj	A treesObject created by readTrees

Value

A trait tree with the correct topology

flat_to_ij	<i>Flat_to_ij</i>
------------	-------------------

Description

Takes a flat matrix index and returns it as two columns or row (i) and column (j) indeces

Usage

```
flat_to_ij(index, ncol)
```

Arguments

index	A flast index
ncol	The number of columns in the matrix

Value

A two column matrix of indeces corresponding to i and j

foreground2Paths	<i>Creates paths from a set of foreground species</i>
------------------	---

Description

Creates paths from a set of foreground species

Usage

```
foreground2Paths(
  foreground,
  treesObj,
  plotTree = F,
  clade = c("ancestral", "terminal", "all", "weighted"),
  useSpecies = NULL
)
```

Arguments

treesObj	A treesObj created by readTrees
plotTree	Plot a tree representation of the result
clade	A character string indicating which branches within the clade containing the foreground species should be set to foreground. Must be one of the strings "ancestral", "terminal", "all", or "weighted".
useSpecies	Give only a subset of the species to use for ancestral state reconstruction (e.g., only those species for which the trait can be reliably determined).
foreground.	A character vector containing the foreground species

Value

A vector of length equal to the number of paths in treesObj

foreground2Tree	<i>Creates a binary trait tree from a set of foreground species.</i>
-----------------	--

Description

Creates a binary trait tree from a set of foreground species.

Usage

```
foreground2Tree(
  foreground,
  treesObj,
  plotTree = T,
  clade = c("ancestral", "terminal", "all"),
  weighted = F,
  transition = "unidirectional",
  useSpecies = NULL
)
```

Arguments

treesObj	A treesObj created by readTrees
plotTree	Plot a tree representation of the result
clade	A character string indicating which branches within the clade containing the foreground species should be set to foreground. Must be one of the strings "ancestral", "terminal", "all".
weighted	if set to TRUE weights foreground edges belonging to the same clade such that their branch lengths sum up to 1 (only done for clade options "all" and "terminal").
transition	A character string indicating whether transitions between background and foreground branches are "bidirectional" or "unidirectional" (no foreground to background transitions, the default)
useSpecies	Give only a subset of the species to use for ancestral state reconstruction (e.g., only those species for which the trait can be reliably determined).
foreground.	A character vector containing the foreground species
collapse2anc	Put all the weight on the ancestral branch when the trait appears on a whole clade (redundant to "clade", kept for backwards compatibility)
wholeClade	Whether to implement the weighted edge option across all members of a foreground clade (redundant to "clade", kept for backwards compatibility)

Value

A tree with edge.lengths representing phenotypic states

foreground2TreeClades	<i>Generates a binary phenotype tree using the list of tip foreground animals, foreground common ancestors, and their phylogenetic relationships</i>
-----------------------	--

Description

Generates a binary phenotype tree using the list of tip foreground animals, foreground common ancestors, and their phylogenetic relationships

Usage

```

foreground2TreeClades(
  fg_vec,
  sisters_list = NULL,
  trees,
  plotTree = T,
  useSpecies = NULL
)

```

Arguments

fg_vec	A vector containing the tip foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
trees	treesObj from readTrees
plotTree	A boolean indicator for plotting the output tree (default=FALSE)
useSpecies	An array containing the tip labels in the output tree

Value

fg.tree A binary phenotype tree corresponding to the input information

generatePermulatedBinPhen

Produces binary permutations for a gene

Description

Produces binary permutations for a gene

Usage

```

generatePermulatedBinPhen(
  tree,
  numperms,
  trees,
  root_sp,
  fg_vec,
  sisters_list,
  pathvec,
  permmode = "cc"
)

```


Arguments

tree	Tree of the gene of interest (if permmode="cc", set this as the masterTree in trees (i.e., the output from readTrees))
numperms	An integer number of permutations
trees	treesObj from readTrees
root_sp	The species to root the tree on
fg_vec	A vector containing the foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
pathvec	A path vector generated from the real set of foreground animals

Value

output.list a list containing the set of binary permulated trees

generatePermulatedBinPhenSSMBatched

Produces binary SSM permutations for a list of genes

Description

Produces binary SSM permutations for a list of genes

Usage

```
generatePermulatedBinPhenSSMBatched(
  trees_list,
  numperms,
  trees,
  root_sp,
  fg_vec,
  sisters_list,
  pathvec,
  permmode = "ssm"
)
```

Arguments

trees_list	A list containing the trees of all genes of interest (formatted like trees in treesObj from readTrees)
numperms	An integer number of permutations
trees	treesObj from readTrees
root_sp	The species to root the tree on
fg_vec	A vector containing the foreground species

sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
pathvec	A path vector generated from the real set of foreground animals
permmod	Mode of binary permutation (Must be "ssm" or "ssmLegacy" for this function; default "ssm")

Value

simPhenoList A list containing binary permulated trees for each gene

getAllCor	<i>Computes the association statistics between RER from getAllResiduals and a phenotype paths vector made with tree2Paths or char2Paths</i>
-----------	---

Description

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype paths vector made with [tree2Paths](#) or [char2Paths](#)

Usage

```
getAllCor(
  RERmat,
  charP,
  method = "auto",
  min.sp = 10,
  min.pos = 2,
  winsorizeRER = NULL,
  winsorizetrait = NULL,
  weighted = F,
  bootstrap = F,
  bootn = 1000,
  sort = F
)
```

Arguments

RERmat	RER matrix returned by getAllResiduals
charP	phenotype vector returned by tree2Paths or char2Paths
method	Method used to compute correlations. Accepts the same arguments as cor .
min.sp	Minimum number of species that must be present for a gene
min.pos	Minimum number of species that must be present in the foreground (non-zero phenotype values)

winsorizeRER	Winsorize RER values before computing Pearson correlation. winsorizeRER=3 will set the 3 most extreme RER values at each end of each row to the value closest to 0.
winsorizetrait	Winsorize trait values before computing Pearson correlation. winsorizetrait=3 will set the 3 most extreme trait values at each end to the value closest to 0.
weighted	perform weighted correlation. This option needs to be set if the clade weights computed in foreground2Tree(wholeClade=T) are to be used. This setting will treat the clade a single observation for the purpose of p-value estimation.
bootstrap	toggle bootstrapping (for weighted pearson correlation)
bootn	number of runs to use when bootstrapping. Will be ignored if bootstrap is false.
sort	If set to TRUE, sorts rows by $\log_{10}(\text{P-value}) * -1 \setminus \text{sign}(\text{Rho})$

Value

A list object with correlation values, p-values, and the number of data points used for each tree

Note

winsorize is in terms of number of observations at each end, NOT quantiles

getAllCorExtantOnly	<i>Computes the association statistics between RER from getAllResiduals and a phenotype vector for phenotype values at the tips of the tree</i>
---------------------	---

Description

Computes the association statistics between RER from [getAllResiduals](#) and a phenotype vector for phenotype values at the tips of the tree

Usage

```
getAllCorExtantOnly(
  RERmat,
  phenvals,
  method = "auto",
  min.sp = 10,
  min.pos = 2,
  winsorizeRER = NULL,
  winsorizetrait = NULL
)
```

Arguments

RERmat	RER matrix returned by getAllResiduals
method	Method used to compute correlations. Accepts the same arguments as cor . Set to "auto" to select automatically based on the number of unique values in charP. This will also auto set the winsorization for Pearson correlation. Set winsorize-trait=some number and winsorizeRER=some number to override
min.sp	Minimum number of species that must be present for a gene
min.pos	Minimum number of species that must be present in the foreground (non-zero phenotype values)
winsorizeRER	Winsorize RER values before computing Pearson correlation. winsorizeRER=3 will set the 3 most extreme RER values at each end of each row to the value closest to 0.
winsorizetrait	Winsorize trait values before computing Pearson correlation. winsorizetrait=3 will set the 3 most extreme trait values at each end to the value closest to 0.

Value

A list object with correlation values, p-values, and the number of data points used for each tree

Note

winsorize is in terms of number of observations at each end, NOT quantiles

getAllResiduals	<i>A wrapper function that runs transformPaths, coreGetResiduals, and getRMat.</i>
-----------------	--

Usage

```
getAllResiduals(
  treesObj,
  transform = "sqrt",
  impute = T,
  nvMod = NULL,
  n.pcs = 0,
  cutoff = NULL,
  useSpecies = NULL,
  min.sp = 10,
  min.valid = 20,
  doOnly = NULL,
  maxT = NULL,
  block.do = F,
  weights = NULL,
  use.weights = T,
  interaction = F,
```

```

    all = F,
    use.rows = NULL,
    norm = "scale"
  )

```

Arguments

<code>treesObj</code>	tree object to operate on, product of readTrees
<code>transform</code>	What transformation to apply. "sqrt" by default.
<code>impute</code>	Whether to impute missing data
<code>nvMod</code>	A normalization model with <code>nrow(treesObj\$paths)</code> rows and any number of columns (optional).
<code>n.pcs</code>	Number of principal components to normalize by (default: 0, mean normalization).
<code>cutoff</code>	A cutoff for branches to be ignored (optional, lowest 5 <code>\itemuseSpeciesSpecies</code> subset to use (optional). <code>\itemmin.sp</code>Minimum species in a tree. (default: 10). <code>\itemmin.valid</code>Minimum number of non NA values (after filtering) that must be present for regression to be computed (default: 20). <code>\itemdoOnly</code>Only do specific trees (optional). <code>\itemmaxT</code>Maximum number of trees to do (optional, mostly useful for debugging). <code>\itemblock.doProcess</code> rows in blocks (default: FALSE). <code>\itemweights</code>Weights for calculations (optional, if <code>treesObj</code> has weights those are used). <code>\itemuse.weights</code>Can be used to turn on/off weighted regression. <code>\iteminteraction</code>Use PC interactions when building the normalization model (default: FALSE). <code>\itemall</code>Logical, indicating whether to return all of the residuals instead of the independent set (default is FALSE). <code>\itemuse.rows</code>A vector of indices of rows/features to return (default is NULL, which means all rows). <code>\itemnorm</code>A character string specifying the normalization method. Options include "scale," "zscore," or "quantile" (default is "scale").

An residual matrix equivalent to that produced by the original [getAllResiduals](#) assuming `all=FALSE`

A wrapper function that runs [transformPaths](#), [coreGetResiduals](#), and [getRMat](#).

getAncLiks	<i>Returns ancestral likelihoods at each node. Based on code from ace in ape and fitMk/rerootingMethod in phytools</i>
------------	--

Description

Returns ancestral likelihoods at each node. Based on code from ace in ape and fitMk/rerootingMethod in phytools

Usage

```
getAncLiks(tree, tipvals, Q = NULL, rate_model = "ER", root_prior = "auto")
```

Arguments

tree	Object of class phylo that has been pruned to only contain the species in tipvals
tipvals	The phenotype data for the extant species in the tree in the same order as tree\$tip.label and mapped to integers
Q	A transition matrix, if NULL the transition matrix is fit with fit_mk from castor package
rate_model	The rate model to use for fitting the transition matrix if one is not provided
root_prior	The root prior used when fitting the transition matrix if one is not provided

Value

The ancestral likelihoods at each node in the tree

getBinaryPermutationInputsFromTree	<i>Produces the phenotype inputs for binary permutation functions</i>
------------------------------------	---

Description

Produces the phenotype inputs for binary permutation functions

Usage

```
getBinaryPermutationInputsFromTree(fgTree)
```

Arguments

fgTree	A binary phenotype tree containing the phenotype values (branch length = 1 for foreground species, 0 for background species)
--------	--

Value

out A list containing fg_vec and sisters_list for binary permutation functions

getEnrichPerms	<i>Performs enrichment statistic permutations using existing gene correlation permutations</i>
----------------	--

Description

Performs enrichment statistic permutations using existing gene correlation permutations

Usage

```
getEnrichPerms(corperms, realenrich, annotlist)
```

Arguments

corperms	Gene correlation permutations from getPermsContinuous
realenrich	Pathway enrichment results using observed phenotype obtained from ‘correlate-WithContinuousPhenotype’ or ‘correlateWithBinaryPhenotype’
annotlist	Pathway annotations

Value

Full null permutation statistics and p-values for gene correlations and pathway enrichment

getForegroundInfoClades	<i>Generates a binary phenotype tree and foreground clades information using the list of tip foreground animals, the presence of foreground common ancestors, and their phylogenetic relationships</i>
-------------------------	--

Description

Generates a binary phenotype tree and foreground clades information using the list of tip foreground animals, the presence of foreground common ancestors, and their phylogenetic relationships

Usage

```
getForegroundInfoClades(
  fg_vec,
  sisters_list = NULL,
  trees,
  plotTree = T,
  useSpecies = NULL
)
```

Arguments

fg_vec	A vector containing the tip foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
trees	treesObj from readTrees
plotTree	A boolean indicator for plotting the output tree (default=FALSE)
useSpecies	An array containing the tip labels in the output tree

Value

output.list A list containing 1) "tree" = a binary phenotype tree corresponding to the input information, 2) "fg.sisters.table" = a table containing all sister species in the foreground set

getLikelihoodRatio	<i>compute the likelihood ratio from the log likeilhoods of two transition matrices</i>
--------------------	---

Description

compute the likelihood ratio from the log likeilhoods of two transition matrices

Usage

```
getLikelihoodRatio(limited, general)
```

Arguments

limited	The log likelihood of the simpler model
general	The log likelihood of the more complex model

Value

the likelihood ratio between two rate models

getLRnullDistribution *Returns the p value of a likelihood ratio based on the simulated null distribution*

Description

Returns the p value of a likelihood ratio based on the simulated null distribution

Usage

```
getLRnullDistribution(  
  LR,  
  tree,  
  Q,  
  null_rm,  
  alt_rm,  
  nsims = 100,  
  plot = FALSE,  
  ...  
)
```

Arguments

LR	the likelihood ratio between two rate models
tree	the phylogenetic tree to obtain the null distribution on
Q	the transition matrix fit on the null rate model
null_rm	the null rate model (the simpler rate model)
alt_rm	the alternative rate model (the more complex rate model)
nsims	the number of simulations to use generate the null distribution
plot	a boolean expressing whether or not to plot a histogram of likelihood ratios representing the null distribution

Value

the p value of a given likelihood ratio based on the null distribution that is generated

getMatrixFromAbbr	<i>Gets the matrix form of a rate model from its abbreviation</i>
-------------------	---

Description

Gets the matrix form of a rate model from its abbreviation

Usage

```
getMatrixFromAbbr(abbr, Nstates)
```

Arguments

abbr	The abbreviation for the rate model. Can be one of "ER", "ARD", or "SYM"
Nstates	The number of phenotype states to include in the rate model

Value

a rate model as an Nstates x Nstates matrix

getPermsBinary	<i>Calculates permuted correlation and enrichment statistics for binary phenotype</i>
----------------	---

Description

Calculates permuted correlation and enrichment statistics for binary phenotype

Usage

```
getPermsBinary(
  numperms,
  fg_vec,
  sisters_list,
  root_sp,
  RERmat,
  trees,
  mastertree,
  permmode = "cc",
  method = "k",
  min.pos = 2,
  trees_list = NULL,
  calculateenrich = F,
  annotlist = NULL
)
```

Arguments

numperms	An integer number of permutations
fg_vec	A vector containing the tip foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
root_sp	The species to root the tree on
RERmat	An RER matrix calculated using getAllResiduals .
trees	treesObj from readTrees
mastertree	A rooted, fully dichotomous tree derived from the treesObj master tree from readTrees . Must not contain species not in traitvec
permmode	Mode of binary permutation ("cc" for Complete Cases (default), "ssm" for Species Subset Match. Old versions of these modes have been moved to ccLegacy and ssmLegacy)
method	statistical method to use for correlations (set to "k" (default) for Kendall Tau test)
min.pos	minimum number of foreground species (default 2)
trees_list	A list containing the trees of all genes of interest (formatted like trees in treesObj from readTrees)
calculateenrich	A boolean variable indicating if null permutation p-values for enrichment statistics
annotlist	Pathway annotations

Value

A list object with enrichment statistics, correlation p-val, rho, and correlation effect size

```
getPermsBinaryExtantOnly
```

Calculates permuted correlation and enrichment statistics for binary phenotype when performing an extant only analysis

Description

Calculates permuted correlation and enrichment statistics for binary phenotype when performing an extant only analysis

Usage

```

getPermsBinaryExtantOnly(
  numperms,
  fg_vec,
  sisters_list,
  root_sp,
  RERmat,
  trees,
  mastertree,
  permmode = "cc",
  method = "k",
  min.pos = 2,
  trees_list = NULL,
  calculateenrich = F,
  annotlist = NULL
)

```

Arguments

numperms	An integer number of permutations
fg_vec	A vector containing the tip foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
root_sp	The species to root the tree on
RERmat	An RER matrix calculated using getAllResiduals .
trees	treesObj from readTrees
mastertree	A rooted, fully dichotomous tree derived from the treesObj master tree from readTrees . Must not contain species not in traitvec
permmode	Mode of binary permutation ("cc" for Complete Cases (default), "ssm" for Species Subset Match)
method	statistical method to use for correlations (set to "k" (default) for Kendall Tau test)
min.pos	minimum number of foreground species (default 2)
trees_list	A list containing the trees of all genes of interest (formatted like trees in treesObj from readTrees)
calculateenrich	A boolean variable indicating if null permutation p-values for enrichment statistics
annotlist	Pathway annotations

Value

A list object with enrichment statistics, correlation p-val, rho, and correlation effect size

getPermsBinaryFudged	<i>Calculates permuted correlation and enrichment statistics for binary phenotype (does not enforce matching structure of sister species, but maintains internal & tips foreground counts within a fudge of the original tree)</i>
----------------------	--

Description

Calculates permuted correlation and enrichment statistics for binary phenotype (does not enforce matching structure of sister species, but maintains internal & tips foreground counts within a fudge of the original tree)

Usage

```
getPermsBinaryFudged(  
  fgdspecs,  
  RERs,  
  trees,  
  useSpecies,  
  ntrees,  
  root,  
  fudge = 5,  
  cors,  
  phenvec  
)
```

Arguments

fgdspecs	A vector containing the tip foreground species
RERs	An RER matrix calculated using getAllResiduals .
trees	treesObj from readTrees
useSpecies	A vector of species to include
ntrees	An integer number of permutations
root	The species to root the tree on
fudge	The number up to which the permulated tree can differ in total foreground species
cors	The gene correlation results from correlateWithBinaryPhenotype
phenvec	A named vector of the phenotype

Value

A list object with enrichment statistics, correlation p-val, rho, and correlation effect size

getPermsContinuous	<i>Calculates permuted correlation and enrichment statistics</i>
--------------------	--

Description

Calculates permuted correlation and enrichment statistics

Usage

```
getPermsContinuous(
  numperms,
  traitvec,
  RERmat,
  annotlist,
  trees,
  mastertree,
  calculateenrich = T,
  type = "simperm",
  winR = 3,
  winT = 3,
  method = "p",
  min.pos = 0
)
```

Arguments

numperms	An integer number of permutations
traitvec	A named phenotype vector
RERmat	An RER matrix calculated using getAllResiduals
annotlist	Pathway annotations
trees	treesObj from readTrees
mastertree	A rooted, fully dichotomous tree derived from the treesObj master tree from readTrees . Must not contain species not in traitvec
calculateenrich	A boolean variable indicating if null permutation p-values for enrichment statistics
type	One of "simperm", "sim", or "perm" for permutations, simulations, or permutations, respectively
winR	Integer winzorization value for RERmat
winT	Integer winzorization value for trait
method	statistical method to use for correlations
min.pos	minimum foreground species - should be set to 0

Value

A list object with enrichment statistics, correlation p-val, rho, and correlation effect size

Note

winsorize is in terms of number of observations at each end, NOT quantiles

`getPermsContinuousExtantOnly`

Calculates permuted correlation and enrichment statistics for an extant only analysis

Description

Calculates permuted correlation and enrichment statistics for an extant only analysis

Usage

```
getPermsContinuousExtantOnly(  
  numperms,  
  traitvec,  
  RERmat,  
  annotlist,  
  trees,  
  mastertree,  
  calculateenrich = T,  
  type = "simperm",  
  winR = 3,  
  winT = 3,  
  method = "p",  
  min.pos = 0  
)
```

Arguments

numperms	An integer number of permutations
traitvec	A named phenotype vector
RERmat	An RER matrix calculated using getAllResiduals
annotlist	Pathway annotations
trees	treesObj from readTrees
mastertree	A rooted, fully dichotomous tree derived from the treesObj master tree from readTrees . Must not contain species not in traitvec
calculateenrich	A boolean variable indicating if null permutation p-values for enrichment statistics

type	One of "simperm", "sim", or "perm" for permutations, simulations, or permutations, respectively
winR	Integer winzorization value for RERmat
winT	Integer winzorization value for trait
method	statistical method to use for correlations
min.pos	minimum foreground species - should be set to 0

Value

A list object with enrichment statistics, correlation p-val, rho, and correlation effect size

Note

winzorize is in terms of number of observations at each end, NOT quantiles

getRMat	<i>Extract a maximal independent residual matrix for each feature Matrix</i>
---------	--

Description

Extract a maximal independent residual matrix for each feature Matrix

Usage

```
getRMat(resOut, all = F, use.rows = NULL, norm = "scale")
```

Arguments

resOut	An object resulting from the NEW coreGetResiduals (NOT the original getAllResiduals)
all	Logical, indicating whether to return all of the residuals instead of the independent set (default is FALSE).
use.rows	A vector of indices of rows/features to return (default is NULL, which means all rows).
norm	A character string specifying the normalization method. Options include "scale," "zscore," or "quantile" (default is "scale").

Value

A residual matrix equivalent to that produced by the original [getAllResiduals](#) assuming all=FALSE

getStat	<i>Calculates Rho-signed negative log-base-ten p-value for use in enrichment functions</i>
---------	--

Description

Calculates Rho-signed negative log-base-ten p-value for use in enrichment functions

Usage

```
getStat(res)
```

Arguments

res	The output from RERconverge correlation functions (correlateWithContinuousPhenotype, correlateWithBinaryPhenotype, getAllCor)
-----	---

Value

A dataframe of Rho-signed negative log-base-ten p-values for all genes, NAs removed

getStatesAtNodes	<i>returns the states at each node corresponding to the state with the max likelihood</i>
------------------	---

Description

returns the states at each node corresponding to the state with the max likelihood

Usage

```
getStatesAtNodes(ancliiks, confidence_threshold = NULL)
```

Arguments

ancliiks	a table of ancestral likelihoods with rows in node order and columns corresponding to phenotype states
confidence_threshold	the default is NULL, but if provided it will only obtain states from nodes whose max likelihood is greater than or equal to the confidence threshold

Value

a vector of states for the internal nodes in order of the internal nodes

getStatesFromPhenTree *Returns a list of states in node order from a phenotype tree*

Description

Returns a list of states in node order from a phenotype tree

Usage

```
getStatesFromPhenTree(phenTree, root_state, tipvals)
```

Arguments

phenTree	The phenotype tree returned by char2TreeCategorical
root_state	The state of the root node, this is not stored along the edge lengths of the tree
tipvals	The states of the tips of the tree in order of phenTree\$tip.label

Value

Returns a vector of the category states of species in order of the nodes in the tree

makeBinaryPaths	<i>Makes a binary path vector from either a tree of class "phylo" or a foreground species set supplied as a character vector</i>
-----------------	--

Description

Makes a binary path vector from either a tree of class "phylo" or a foreground species set supplied as a character vector

Usage

```
makeBinaryPaths(input, treesObj)
```

Arguments

input	Either a phenotype tree of class "phylo" (with branch length encoding a phenotype) or a character vector of foreground branches
treesObj	A treesObj created by readTrees
transition	A character string indicating whether transitions between background and foreground branches are "bidirectional" or "unidirectional" (no foreground to background transitions, the default)

Value

A vector of length equal to the number of paths in treesObj

`mapIndex`*Map index*

Description

Creates a map from the paths in one tree object to the paths in the other object Can be used to put old style and new style trees objects into the same order

Usage

```
mapIndex(treesObj1, treesObj2)
```

Arguments

<code>treesObj1</code>	The first trees object
<code>treesObj2</code>	The second trees object

Value

The index of the paths matrix columns in treesObj1 inside treesObj2

`matchAllNodesClades`*Calculates the clade mappings between the gene tree and the master tree (with the complete topology)*

Description

Calculates the clade mappings between the gene tree and the master tree (with the complete topology)

Usage

```
matchAllNodesClades(gene_tree, treesObj)
```

Arguments

<code>gene_tree</code>	A binary phenotype tree of a gene
<code>treesObj</code>	treesObj from readTrees

Value

`output.map` A list containing a dataframe of clades mapping

namePathsWSpecies	<i>Provides names for paths/RERs representing terminal branches for plotting Originally an internal function but necessary for the vignette/walk-through</i>
-------------------	--

Description

Provides names for paths/RERs representing terminal branches for plotting Originally an internal function but necessary for the vignette/walk-through

Usage

```
namePathsWSpecies(masterTree)
```

Arguments

masterTree	The master tree used for analysis
------------	-----------------------------------

Value

Names corresponding to the paths/RERs for terminal branches

permpvalcor	<i>Calculates correlation permutation pvals from output of getPermsContinuous and getPermsBinary</i>
-------------	--

Description

Calculates correlation permutation pvals from output of [getPermsContinuous](#) and [getPermsBinary](#)

Usage

```
permpvalcor(realcor, permvals)
```

Arguments

realcor	Real enrichment statistics from fastwilcoxGMTall
permvals	output from getPermsContinuous or getPermsBinary

Value

A data frame containing permutation p-values and permutation statistics (positive denotes acceleration, negative denotes deceleration)

permpvalenrich	<i>Calculates enrichment permutation pvals from output of getPermsContinuous</i>
----------------	--

Description

Calculates enrichment permutation pvals from output of [getPermsContinuous](#)

Usage

```
permpvalenrich(realenrich, permvals)
```

Arguments

realenrich	Real enrichment statistics from fastwilcoxGMTall
permvals	output from getPermsContinuous

Value

A list object with vectors of permutation p-values

permutevec	<i>Generates a permuted continuous phenotype given an observed continuous phenotype</i>
------------	---

Description

Generates a permuted continuous phenotype given an observed continuous phenotype

Usage

```
permutevec(namedvec)
```

Arguments

namedvec	A named numeric vector with phenotype values for each speices
----------	---

Value

A vector with permuted phenotype values

```
plotPositivesFromPermutations
```

Plots changes in number of statistically significant rate acceleration or deceleration versus the number of permutations

Description

Plots changes in number of statistically significant rate acceleration or deceleration versus the number of permutations

Usage

```
plotPositivesFromPermutations(
  res,
  perm.out,
  interval,
  pvalthres,
  output.res = FALSE
)
```

Arguments

<code>res</code>	correlation statistic output from correlateWithBinaryPhenotype or correlateWithContinuousPhenotype
<code>perm.out</code>	output from getPermsBinary or getPermsContinuous
<code>interval</code>	interval of number of permutations (e.g., interval = 10 means that number of positives will be calculated for number of permutations = 10, 20, 30, ...)
<code>pvalthres</code>	p-value threshold for identifying statistically significant rate acceleration or deceleration
<code>output.res</code>	Boolean defining whether to output a list object containing changes in the number of identified elements with changing number of permutations (default=FALSE)

Value

A list containing changes in the number of identified elements with changing number of permutations (default=NULL)

<code>plotRers</code>	<i>Plot the residuals reflecting the relative evolutionary rates (RERs) of a gene across species present in the gene tree</i>
-----------------------	---

Description

Plot the residuals reflecting the relative evolutionary rates (RERs) of a gene across species present in the gene tree

Usage

```
plotRers(
  rermat = NULL,
  index = NULL,
  phenv = NULL,
  rers = NULL,
  species_from = NULL,
  species_to = NULL,
  method = "k",
  xlims = NULL,
  plot = 1,
  xextend = 0.2,
  sortrers = F
)
```

Arguments

rermat. A residual matrix, output of the `getAllResiduals()` function

index. A character denoting the name of gene, or a numeric value corresponding to the gene's row index in the residuals matrix

phenv. A phenotype vector returned by [tree2Paths](#) or [foreground2Paths](#)

species_from/species_to. Name translations: two vectors of equal length with corresponding elements. The function will translate any occurrence of `species_from[n]` to `species_to[n]` (if applicable).

Value

A plot of the RERs with foreground species labelled in red, and the rest in blue

plotTreeCategorical	<i>Plot a phenotype tree generated by char2TreeCategorical with the branches colored by category</i>
---------------------	--

Description

Plot a phenotype tree generated by [char2TreeCategorical](#) with the branches colored by category

Usage

```
plotTreeCategorical(
  tree,
  category_names = NULL,
  master = NULL,
  node_states = NULL
)
```

Arguments

tree	The phenotype tree returned by char2TreeCategorical
category_names	The names of the categories in order of the corresponding numerical labels
master	If provided, the tree will be plotted using the branch lengths of that tree, coloring branches by the phenotype in the phenotype tree. The master tree and phenotype tree should have the same topology and the same species. If not provided, use.edge.length is set to FALSE.
node_states	A vector of the phenotype state at each node in the tree (in order of the nodes). If provided, vertical lines representing nodes in the tree are colored by state. It must be numerical and the integers must have the same mapping to the category names as in the phenotype tree.

Value

A plot of a phylogenetic tree with branches colored by phenotype category.

plotTreeHighlightBranches

Plot the provided tree, (optionally) rerooted, with specified branches highlighted

Description

Plot the provided tree, (optionally) rerooted, with specified branches highlighted

Usage

```
plotTreeHighlightBranches(
  tree,
  outgroup = NULL,
  hlspecies,
  hlcols = NULL,
  main = ""
)

plotTreeHighlightBranches(
  tree,
  outgroup = NULL,
  hlspecies,
  hlcols = NULL,
  main = ""
)
```


Arguments

hlspecies	the species to highlight
hlcols	the colors to highlight those species with
main	the name to give the plot
tree.	A tree object.
outgroup.	A vector of species to use to root the tree. If not provided, the tree remains unrooted.
hlspecies.	A vector of species whose terminal branches to highlight, or a vector of branch numbers within the tree.
hlcols.	Colors to use in highlighting the branches. If not specified, will use default R colors.
main.	Main text for plot.
useGG	option to creat plot with ggtree. Improves plot readability for most data sets.

Value

A plot of the the (optionally rerooted) tree, with branches highlighted.

plotWithCor	<i>Scatter plot with correlation</i>
-------------	--------------------------------------

Description

Scatter plot with correlation

Usage

```
plotWithCor(x, y)
```

pruneTree	<i>wrapper around drop.tip</i>
-----------	--

Description

wrapper around [drop.tip](#)

Usage

```
pruneTree(tree, tip.names)
```

Arguments

tree	a "phylo" tree
tip.names	The tip names to keep in the tree

Value

A new pruned tree

read.gmt	<i>Reads pathway data from a gmt-formatted file from GSEA-MSigDB</i>
----------	--

Description

Reads pathway data from a gmt-formatted file from GSEA-MSigDB

Usage

```
read.gmt(filename)
```

Arguments

filename The gmt-formatted pathway file

Value

A list of pathway information with pathway names and genesets

readTrees	<i>Reads trees from a 2 column , tab seperated, file The first columns is the gene name and the second column is the corresponding tree in parenthetic format known as the Newick or New Hampshire format This function is faster than readTrees but returns paths in a different indexing order</i>
-----------	--

Description

Reads trees from a 2 column , tab seperated, file The first columns is the gene name and the second column is the corresponding tree in parenthetic format known as the Newick or New Hampshire format This function is faster than readTrees but returns paths in a different indexing order

Usage

```
readTrees(  
  file,  
  max.read = NA,  
  masterTree = NULL,  
  minTreesAll = 20,  
  reestimateBranches = F,  
  minSpecs = NULL,  
  useSpecies = NULL  
)
```

Arguments

file	The path to the tree file
max.read	This function can take several minutes, so max.read is useful for testing
masterTree	(optional) User can specify a master tree; only the topology will be used, and branch lengths will be inferred from gene trees. Recommended only when the number of available gene trees with all species is small.
minTreesAll	The minimum number of trees with all species present in order to estimate master tree edge lengths (default 20).
reestimateBranches	Boolean indicating whether to re-estimate branch lengths if master tree topology is included (default FALSE)
minSpecs	the minimum number of species that needs to be present in a gene tree to be included in calculating master tree
useSpecies	Species subset to use (optional).

Value

A trees object of class "treeObj"

RERconverge	<i>RERconverge</i>
-------------	--------------------

Description

RERconverge
RERconverge
RERconverge

returnRersAsNewickStrings	<i>Produce a vector of newick strings representing gene trees where the edge lengths correspond to RERs</i>
---------------------------	---

Description

Produce a vector of newick strings representing gene trees where the edge lengths correspond to RERs

Usage

```
returnRersAsNewickStrings(treesObj, rermat)
```

Arguments

treesObj. A treesObj created by [readTrees](#)
 rermat. A residual matrix, output of the getAllResiduals() function

Value

A named character vector of newick strings, one per gene, representing RERs as edge lengths'

returnRersAsTree	<i>Produce a gene tree with branch lengths representing RERs and (optionally) display these RERs as branch labels</i>
------------------	---

Description

Produce a gene tree with branch lengths representing RERs and (optionally) display these RERs as branch labels

Usage

```

returnRersAsTree(
  treesObj,
  rermat,
  index,
  phenv = NULL,
  rer.cex = 0.7,
  tip.cex = 0.7,
  nalab = "NA",
  plot = T
)

```

Arguments

treesObj. A treesObj created by [readTrees](#)
 rermat. A residual matrix, output of the getAllResiduals() function
 index. A character denoting the name of gene, or a numeric value corresponding to the gene's row index in the residuals matrix
 phenv. A phenotype vector returned by [tree2Paths](#) or [foreground2Paths](#)
 rer.cex. Numeric expansion for RER labels
 tip.cex. Numeric expansion for tip labels
 nalab. Label given to any NA RERs
 plot. Whether to produce a plot displaying the RERs on the gene tree

Value

An object of class "phylo" with edge lengths representing RERs for the given gene
 If plot = TRUE, also displays a plot of the gene tree with edges labeled with RERs

returnRersAsTreesAll	<i>Produce a multiPhylo object of all gene trees with branch lengths representing RERs for each gene</i>
----------------------	--

Description

Produce a multiPhylo object of all gene trees with branch lengths representing RERs for each gene

Usage

```
returnRersAsTreesAll(treesObj, rermat)
```

Arguments

treesObj.	A treesObj created by readTrees
rermat.	A residual matrix, output of the <code>getAllResiduals()</code> function

Value

An object of class "multiPhylo" of named gene trees with edge lengths representing RERs for the given gene

searchRateModels	<i>iteratively searches for simpler rate models that still provide a good fit to the observed data at the tips of the tree</i>
------------------	--

Description

iteratively searches for simpler rate models that still provide a good fit to the observed data at the tips of the tree

Usage

```
searchRateModels(
  treesObj,
  phenvals,
  pthreshold = 0.05,
  lthreshold = 4,
  max_iterations = 2000,
  ...
)
```

Arguments

treesObj	the trees object returned by readTrees
phenvals	the named phenotype vector with names matching the tip labels in the trees object
pthreshold	the more complex model is considered significantly better when the p value is less than pthrehsold
lthreshold	the more compelx model is considered significantly better when the likelihood ratio is greater than the lthreshold, used when the more complex and simpler models are not nested
max_iterations	stops the algorithm if the number of iterations exceeds max_iterations even if it isn't done in order to avoid extremely long runtimes
...	additional parameters ot pass to fit_mk, the castor function used to fit a transition matrix given a rate model, phylogenetic tree, and phenotype data

Value

returns a list of the rate models generated during the iterative search for simpler rate models and a table of statistics including the likelihood ratios computed on each iteration

simBinPhenoCC	<i>Produces one CC binary permutation for a gene</i>
---------------	--

Description

Produces one CC binary permutation for a gene

Usage

```
simBinPhenoCC(
  trees,
  mastertree,
  root_sp,
  fg_vec,
  sisters_list = NULL,
  pathvec,
  plotTreeBool = F
)
```

Arguments

trees	treesObj from readTrees
mastertree	A rooted, fully dichotomous tree derived from the treesObj master tree from readTrees .
root_sp	The species to root the tree on

fg_vec	a vector containing the foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
pathvec	A path vector generated from the real set of foreground animals
plotTreeBool	Boolean indicator for plotting the output tree (default=FALSE)

Value

A CC binary permulated tree

simBinPhenoCCmidpoint *Produces one CC binary permutation for a gene using midpoint rooting*

Description

Produces one CC binary permutation for a gene using midpoint rooting

Usage

```
simBinPhenoCCmidpoint(
  trees,
  mastertree,
  fg_vec,
  sisters_list = NULL,
  pathvec,
  plotTreeBool = F
)
```

Arguments

trees	treesObj from readTrees
mastertree	A rooted, fully dichotomous tree derived from the treesObj master tree from readTrees .
fg_vec	a vector containing the foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
pathvec	A path vector generated from the real set of foreground animals
plotTreeBool	Boolean indicator for plotting the output tree (default=FALSE)

Value

A CC binary permulated tree

simBinPhenoRank	<i>Produces one binary permutation based on ranking of simulated branch lengths</i>
-----------------	---

Description

Produces one binary permutation based on ranking of simulated branch lengths

Usage

```
simBinPhenoRank(trees, root_sp, fg_vec, sisters_list = NULL, plotTreeBool = F)
```

Arguments

trees	treesObj from readTrees
root_sp	The species to root the tree on
fg_vec	a vector containing the foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
plotTreeBool	Boolean indicator for plotting the output tree (default=FALSE)

Value

A binary permulated tree

simBinPhenoSSM	<i>Produces one SSM binary permutation for a gene</i>
----------------	---

Description

Produces one SSM binary permutation for a gene

Usage

```
simBinPhenoSSM(tree, trees, fg_vec, pathvec, plotTreeBool = F)
```

Arguments

tree	Tree of the gene of interest
trees	treesObj from readTrees
fg_vec	A vector containing the foreground species
pathvec	A path vector generated from the real set of foreground animals
plotTreeBool	Boolean indicator for plotting the output tree (default=FALSE)

Value

A SSM binary permulated tree

simBinPhenoSSMLegacy *Produces one SSM binary permutation for a gene*

Description

Produces one SSM binary permutation for a gene

Usage

```
simBinPhenoSSMLegacy(  
  tree,  
  trees,  
  root_sp,  
  fg_vec,  
  sisters_list = NULL,  
  pathvec,  
  plotTreeBool = F  
)
```

Arguments

tree	Tree of the gene of interest
trees	treesObj from readTrees
root_sp	The species to root the tree on
fg_vec	A vector containing the foreground species
sisters_list	A list containing pairs of "sister species" in the foreground set (put NULL if empty)
pathvec	A path vector generated from the real set of foreground animals
plotTreeBool	Boolean indicator for plotting the output tree (default=FALSE)

Value

A SSM binary permulated tree

simpermvec	<i>Generates a permulated continuous phenotype given an observed continuous phenotype and a phylogeny</i>
------------	---

Description

Generates a permulated continuous phenotype given an observed continuous phenotype and a phylogeny

Usage

```
simpermvec(namedvec, treewithbranchlengths)
```

Arguments

namedvec	A named numeric vector with phenotype values for each speices
treewithbranchlengths	A rooted phylogenetic tree with the same species as namedvec and branch lengths representing average evolutionary rate. The master tree from readTrees may be rooted and used for this parameter.

Value

A vector with permulated phenotype values

simulatevec	<i>Generates a simulated continuous phenotype given an observed continuous phenotype and a phylogeny</i>
-------------	--

Description

Generates a simulated continuous phenotype given an observed continuous phenotype and a phylogeny

Usage

```
simulatevec(namedvec, treewithbranchlengths)
```

Arguments

namedvec	A named numeric vector with phenotype values for each speices
treewithbranchlengths	A rooted phylogenetic tree with the same species as namedvec and branch lengths representing average evolutionary rate. The master tree from readTrees may be rooted and used for this parameter.

Value

A vector with simulated phenotype values

transformPaths	<i>Apply a transformation to a trees object</i>
----------------	---

Description

Apply a transformation to a trees object

Usage

```
transformPaths(treesObj, transform = "sqrt", impute = T)
```

Arguments

transform	What transformation to apply. "sqrt" by default.
impute	Whether to impute missing data

Value

A treesObj with transformed paths

tree2Paths	<i>Generate a phenotype paths vector from a phenotype tree</i>
------------	--

Description

tree2Paths generates a phenotype paths vector matching the treesObject from a tree where branches specify phenotypes.

Usage

```
tree2Paths(tree, treesObj, binarize = NULL, useSpecies = NULL, categorical = F)
```

Arguments

tree	A phenotype tree, with branch length encoding a phenotype.
treesObj	A treesObject created by readTrees
binarize	Force binary path representation. Default action depends upon the type of data within the phenotype tree (binary or continuous). • If binary (all branch lengths == 0 or 1): Sets all positive path values to 1. Useful if the tree has non-zero branch lengths for an internal branch or branches; otherwise, values are simply added along branches when calculating paths. Default behavior: binarize = TRUE.

- If continuous (not all branch lengths == 0 or 1): Sets all path values > the mean to 1 and all those <= the mean to 0. Converts a continuous phenotype to a binary phenotype, with state determined by comparison to the mean across all paths. Default behavior: binarize = FALSE.
- useSpecies Give only a subset of the species to use for ancestral state reconstruction (e.g., only those species for which the trait can be reliably determined).

Details

The tree topology of the phenotype tree must match that of the master tree within the treesObject.

Value

A vector of length equal to the number of paths in treesObj

tree2PathsClades	<i>A modification of the tree2Paths function that takes in pre-calculated mappings</i>
------------------	--

Description

A modification of the tree2Paths function that takes in pre-calculated mappings

Usage

tree2PathsClades(tree, trees)

Arguments

- tree the input tree to be converted into paths
- trees treesObj from [readTrees](#)

treePlotNew	<i>Plot ‘tree’ with branch labels colored in a heatmap based on values in ‘vals’</i>
-------------	--

Description

Plot ‘tree’ with branch labels colored in a heatmap based on values in ‘vals’

Usage

```

treePlotNew(
  tree,
  vals = NULL,
  rank = F,
  nlevels = 9,
  type = "c",
  col = NULL,
  maintitle = NULL,
  useedge = F,
  doreroot = F,
  rerootby = NULL,
  useSpecies = NULL,
  species.names = NULL,
  speclist1 = NULL,
  speclist2 = NULL,
  aligntip = F,
  colpan1 = rgb(0, 119, 187, maxColorValue = 255),
  colpan2 = rgb(204, 51, 17, maxColorValue = 255),
  colpanmid = rgb(187, 187, 187, maxColorValue = 255),
  plotspecies = NULL,
  edgetype = NULL,
  textsize = 0.6,
  colbarlab = "",
  splist2sym = "psi",
  dolegend = T,
  nacol = rgb(0, 0, 0),
  figwid = 10,
  ...
)

```

Arguments

tree.	A phylo object, used for topology
vals.	Values to use for heatmap to color branches, in same order as edges of tree
rank.	Whether to plot ranks instead of values
nlevels.	How many colors to use in the heatmap
type.	Type for plot.phylo
col.	Vector of user-defined colors
maintitle.	Main title label
useedge.	Whether to use edge lengths in 'tree' for plotting
doreroot.	Whether to re-root the tree before plotting
rerootby.	If re-rooting, what to use to root the tree (verify by checking against unrooted plot)
useSpecies.	A vector of species to include in the plot (verify by checking against full set)

species.names.	Data fram for converting names in ‘tree’ to names to be plotted (row names should be tip labels of tree, and first column should contain the corresponding desired tip labels)
speclist1.	A vector of tip labels to highlight in bold, blue text
speclist2.	A vector of tip labels to which to add an asterisk
aligntip.	Whether to align tip labels (default FALSE)
colpan1.	Color for lowest value in heatmap (default blue)
colpan2.	Color for highest value in heatmap (default rose)
colpanmid.	Color for middle value in heatmap (default gray)
plotspecies.	A vector of tip labels to display on the tree (the remainder will be masked, but the corresponding tips will be plotted on the tree)
edgetype.	Vector of line type for edges.
textsize.	cex value for tip labels (default 0.6)
colbarlab.	Label for the color bar legend
splist2sym.	A value within species names to display as a symbol
dolegend.	Whether to display the heatmap legend.
nacol.	Color to display for any edges with length NA
figwid.	Adjust x limits of plot.phylo by 1/figwid. May be related to figure width (requires some optimization)
....	further arguments to be passed to ‘plot’ or to ‘plot.phylo’

Value

Plots a cladogram of ‘tree’ with branch colors determined by ‘vals’

treePlotRers	<i>Plot a cladogram with RERs shown as either labels (type="label") or a color heatmap along the branches (type="color") Wraps around returnRersAsTree or treePlotNew, respectively</i>
--------------	---

Description

Plot a cladogram with RERs shown as either labels (type="label") or a color heatmap along the branches (type="color") Wraps around [returnRersAsTree](#) or [treePlotNew](#), respectively

Usage

```
treePlotRers(
  treesObj,
  rermat = NULL,
  index = NULL,
  type = c("label", "color"),
  phenv = NULL,
  figwid = 10,
  ...
)
```

Arguments

treesObj.	A treesObj created by readTrees
rermat.	A residual matrix, output of the getAllResiduals() function
index.	A character denoting the name of gene, or a numeric value corresponding to the gene's row index in the residuals matrix
type.	Whether to display RERs as branch labels ('label') or a heatmap ('color')
phenv.	A phenotype vector returned by tree2Paths or foreground2Paths
....	Additional parameters to be passed to returnRersAsTree or treePlotNew #'@param figwid. Adjust x limits of plot.phylo by 1/figwid. May be related to figure width (requires some optimization)

Value

Plots a cladogram of the master tree with RERs displayed as branch labels or colors

visCompareTwoRateModels

Visually compare the results of two rate models by plotting the differences between them on the phylogenetic tree Relative entropy is calculated with A as the "target" and B as the "estimate" Switching the order of rate model A and rate model B DOES give different relative entropies (relative entropy is not symmetric)

Description

Visually compare the results of two rate models by plotting the differences between them on the phylogenetic tree Relative entropy is calculated with A as the "target" and B as the "estimate" Switching the order of rate model A and rate model B DOES give different relative entropies (relative entropy is not symmetric)

Usage

```
visCompareTwoRateModels(
  A,
  B,
  treesObj,
  phenvals,
  mode = "entropy",
  cex = 0.5,
  ...
)
```

Arguments

A	the first rate model
B	teh second rate model
treesObj	the trees object returned by readTrees
phenvals	the named phenotype vector with names matching the tip labels in the trees object
mode	indicates how to represent differences between ancestral reconstructions. When mode is "entropy" the relative entropies between ancestral likelihoods at each node are plotted along the branches of the trees. When mode is "match", nodes are colored red when state assignments are different and red when they are the same.
cex	a graphical argument representing the size of node labels and tip labels

Value

a list of relative entropies or state assignemtns and ancestral likelihoods under each rate model

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